

Harik Sodhi

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EDUCATION

University of Oxford (Christ Church) | MEng Engineering Science **Oct 2025 – Jun 2029**

- Relevant modules: Engineering Mathematics, Structures and Mechanics, Electrical and Information Engineering, Energy.

AWARDS & HONOURS

- Silver Medal, International Physics Olympiad (49th/420 competitors from 89 countries; highest-ranked competitor from Western Europe) 2025
- United Kingdom International Physics Olympiad Team Member 2025
- Purple Comet Mathematics Competition — UK National Champion 2024
- Winner, Berkeley Math Tournament Online Power Round 2024
- British Mathematical Olympiad Round 1 Distinction 2024
- Ritange Winner out of 1200 teams (Team Captain) 2023

EXPERIENCE

Oxford Alpha Fund **Oxford**

Quant Analyst / Director *Jan 2026 – Present*

- Selected through a competitive quantitative research and presentation process as a first-year undergraduate.
- Researched volatility risk premia in short-dated Bitcoin options using historical Deribit/Coinbase data; developed and backtested systematic trading strategies achieving a Sharpe ratio of 1.18 and a 70% win rate. [Code]
- Built automated tooling to generate sponsor-ready CV books from member CVs, streamlining recruitment preparation.
- Coordinated with sponsoring firms and industry partners to support recruitment events and strengthen fund relationships.

Optiver Trading Academy **Oxford**

Participant *Oct 2025 – Nov 2025*

- Selected for Optiver's competitive Trading Academy programme as a first-year undergraduate.
- Traded ETFs and futures on Optiver's proprietary simulation platform, applying market-making, delta-hedging, and risk-management principles under realistic market conditions.

LEADERSHIP & ACTIVITIES

VEX Robotics & REC Foundation **London / Remote**

Global Student Advisory Board Member & Team Captain *May 2021 – Mar 2024*

- Selected as one of 12 members globally to directly advise the CEO and executive leadership team on the strategic direction and accessibility of the VEX Robotics programme.
- Served as Team Captain, Lead Programmer, and Lead Designer; managed team operations and spearheaded end-to-end development of competitive robots.
- Engineered autonomous routines and driver-assist features in C++ (PROS), implementing closed-loop control systems (PID) for precise chassis positioning and kinematic sub-systems.
- Utilized 3D CAD software to design structurally optimized mechanisms, translating game constraints into high-performance hardware.

PROJECTS

Market Making Simulator

- Developed a multi-agent market simulation modelling market makers, noise traders, and informed traders.
- Analysed the effects of participant composition on spread formation, adverse selection, and trading profitability.
- Implemented strategy evaluation and performance analysis tools for comparing trading behaviour. [GitHub]

Rocket Lander

- Designed and developed a MATLAB simulation of autonomous rocket landing dynamics from first principles, incorporating variable-mass propulsion, atmospheric drag, and gravity.
- Implemented Runge-Kutta numerical integration, a scratch-built PID control system, and constrained optimisation techniques to achieve stable autonomous touchdowns. [GitHub]

SKILLS

- **Programming Languages:** Python, MATLAB, LaTeX, C++
- **Libraries & Frameworks:** NumPy, Pandas, Matplotlib, Tkinter, PyTorch, Scikit-learn, PyPDF2, IPython
- **Developer Tools:** Git, GitHub